

Вариант 8

Решение уравнения имеет вид

$$\mathbf{U}_{\xi^0} = \frac{1}{2\varepsilon^2} \left[\frac{\partial^2 P_0}{\partial \xi \partial \tau} + \frac{\partial^2 P_0}{\partial \xi \partial \tau} \bar{\Psi}(\zeta) + \frac{\partial P_0}{\partial \xi} \bar{\Phi}(\zeta) \right], \quad (1)$$

где

$$\bar{\Psi}(\zeta) = F_2(\varepsilon\zeta)D_1 - F_1(\varepsilon\zeta) - 2F_4(\varepsilon\zeta)D_2, \quad \bar{\Phi}(\zeta) = 2F_3(\varepsilon\zeta) - F_2(\varepsilon\zeta)D_2 - 2F_4(\varepsilon\zeta)D_1,$$

$$\bar{\Psi}_1(\zeta) = \int_{\zeta}^1 \bar{\Psi} d\zeta, \quad \bar{\Phi}_1(\zeta) = \int_{\zeta}^1 \bar{\Phi}(\zeta) d\zeta, \quad \varepsilon(\omega) = \sqrt{\Re/2},$$

$$F_1(\varepsilon\zeta) = \operatorname{ch} \varepsilon\zeta \cos \varepsilon\zeta, \quad F_2(\varepsilon\zeta) = [\operatorname{ch} \varepsilon\zeta \sin \varepsilon\zeta + \operatorname{sh} \varepsilon\zeta \cos \varepsilon\zeta]/2,$$

$$\gamma(\omega) = \frac{1}{6} \frac{\varepsilon^3 (\operatorname{sh} \varepsilon - \sin \varepsilon)}{\varepsilon^2 (\operatorname{ch} \varepsilon + \cos \varepsilon) - 2\varepsilon (\operatorname{sh} \varepsilon + \sin \varepsilon) + 2(\operatorname{ch} \varepsilon - \cos \varepsilon)}.$$